HIGH FREQUENCY (HF) E-FIELD METER
STATEMENT OF WORK (SOW)

1.0 INTRODUCTION

The requiring activity is the Naval Surface Warfare Center, Dahlgren Division (NSWCDD). NSWDD performs Radiation Hazard (RADHAZ) Survey measurements and electromagnetic environmental effects [redacted] to ensure warfighter safety and capability. These measurements require accurate capture and transmission of frequency and amplitude of radio frequency (RF) signals. This new instrument will replace a High Frequency (HF) Meter approaching the end of life.

2.0 MINIMUM PERFORMANCE SPECIFICATIONS

2.1 The Contractor shall use commercially-off-the-shelf (COTS) products to provide the proposed equipment which must meet the following minimum hardware specifications:

Electrical	Requirement	Unit	Objective	Threshold	Notes
Electrical	Frequency Range	MHz	1.5-30	2-30	High Frequency only
Electrical	Frequency Resolution (steps)	kHz	1	10	Discrete Freq. Measurements
Electrical	Minimum Operational E-field	V/m	0.05	0.1	Limit for accurate measurements
Electrical	Maximum Operational E-field	V/m	400	250	Limit for accurate measurements
Electrical	Maximum Survivability E-field	V/m	1000	700	Limit before it breaks
Electrical	Modulation	-	CW	CW	Only need to calibrate CW for 35-foot antennas on GP
Electrical	E-field Resolution	V/m	0.01	0.1	
Electrical	Measurement Precision (repeatability)	+/- dB	0.1	0.5	
Electrical	Measurement Accuracy	dB	1	3	
Electrical	Battery Operating Time	hours	12	6	Rechargeable (operate system while charging) and/or replaceable and commercially available
Physical	Operating Temperature	F	14 to 130	20 to 110	
Physical	Antenna Ports	-	-	-	Proprietary connection or N-Type, SMA
Physical	Charger	-	-	-	Include a battery charger with equipment. Supports Fast charging.
Physical	Environmental Protection	-	-	-	Water resistance, shipping survivable, shipboard environment ready
Physical	User Interface		Selectors and Display	Selectors and Display	Push buttons
Physical	Non-operating Temperature	F	-30 to 160	-20 to 150	

Physical	Dimensions		Typical Handheld		Smallest size that meet all requirements
Physical	Weight	lb	no more than 2	no more than 5	Able to hold it for 10 mins
Data	Data Port		usb-c	-	for pre/post data load/unload
Data	Calibration Requirements			-	ISO 17025 +Uncertainty Bands +Guardbanding
Electrical	Storage Capacity	GB	>1	>0.5	non-volatile, overwritable while performing measurements

2.2 The proposed equipment must meet the following minimum software specifications:

Category	Requirement	Objective	Threshold	Notes
User Interface	Application Programming Interface	(Integration between E3TO and HF meter)	-	JavaScript Object Notation (JSON) which allows easily readable language between E3 and handheld device. This will allow data from E3 to be written in JSON and parse into JavaScript or Python file.
User Interface	User Display Menu	B55 Shipboard	-	User Display must have: 1. Antenna Number (1-100) 2. Frequency Selection 3. Test Point Location (1-1000) 4. Battery Level
User Interface	Read/Write Formats	Must read/write in xml, csv & txt	-	
User Interface	Max hold	-	-	
User Interface	Clear/Refresh trace	Clear at new test point location	-	
User Interface	Add Trace	-	-	
User Interface	Display Size	Sunlight and nighttime readable, wide enough to include all menu options from row number 2	Sunlight and nighttime readable	Anti-glare

3.0 REQUIREMENTS

3.1 The Contractor shall provide a product that meets long term needs for reliability, maintainability, ease of operation and piece part support.

NOTE: Accordingly, it is intended that the Contractor will provide a commercial off-the-shelf (COTS) or modified COTS system.

3.2 The Contractor shall deliver a total of 10 compliant units.

3.3 The Contractor shall adhere to the following schedule:

Item #	Description	Quantity	Delivery Date
0001	First Article Test (FAT)	1	8 Months After Receipt of Order (ARO)
0002	HF Meter	9	2 Months after receiving FAT Approval
0003	Shipping	1	
0004	Optional – HF Meter	10	10 Months After Options Exercise (AOE)

3.4 The Contractor shall fabricate one (1) HF E-Field Meter according to the minimum performance specifications provided in Section 2.0 prior to fabrication of any full production run.

3.4.1 The Contractor shall deliver the first article quantity eight (8) months after receipt of the order.

NOTE: The Government will perform testing on the first article unit.

3.4.2 The Contractor shall furnish a corrected First Article unit at no additional cost to the Government for subsequent testing if the first article fails any testing or the visual inspection. The Government will notify the Contractor seven (7) calendar days after receipt of the First Article, in writing, of the approval or disapproval of the First Article. The notice of approval or disapproval shall not relieve the contractor from complying with all requirements of the specifications and all other terms and conditions of this order. A notice of disapproval shall cite reasons for disapproval. The Contractor shall make any necessary changes, modification, or repairs to the First Article or select another First Article for testing. All costs related to Contractor performed tests are to be born by the contractor, including all costs for additional tests following a disapproval. If the First Article is disapproved, the Contractor shall provide a pre-paid request for authorization (RMA) for returned shipping costs so the unit may be returned for correction, repair or replacement. Any repaired units must be clearly identified when returned to the Government for subsequent inspection and acceptance testing.

3.4.3 The Contractor shall deliver nine (9) HF E-Field Meter (also referred to as a HF Meter) that meets the minimum performance specifications identified in Section 2.0 after approval of FAT from the Government. The final product shall be easily re-producible.

3.4.4 The Contractor shall email shipment tracking information to the Technical Point of Contact (TPOC) on the day the units are shipped.

3.4.5 The Contractor shall ensure the units provides testing capabilities and environmental characterizations for shore and shipboard surveys.

3.4.6 The Contractor shall provide reports in accordance with (IAW) CDRL A002.

3.5 Capability Demonstration and Testing: The Contractor shall perform acceptance test/functional checkout before the FAT unit is shipped.

NOTE: The Government will witness the test/functional checkout process.

3.5.1 The event can be performed at the Contractor's site or the Government's site. The determination of the time, date and location shall be coordinated with the TPOC identified below at least 30 days prior to the scheduled event.

3.6 The Contractor shall provide onsite training on the fundamentals and operations of the HF Meter.

3.6.1 The Contractor shall provide hands-on training regarding the use and maintenance of the hardware.

3.6.2 The Contractor shall provide technical documentation IAW CDRL A001.

3.6.3 The Contractor shall coordinate with the TPOC identified below to schedule the training during the period of performance of the contract.

3.7 The Contractor shall provide option pricing to produce 10 additional units.

NOTE: The Government reserves the right to waive any optional first article requirements.

*The Contractor shall not fabricate and deliver any additional optional units until the Government applies the appropriate funding to exercise the option and the Contracting Officer provides a subsequent contract modification..

*The Government reserves the right to modify this contract at a later date to cover additional repair services/components identified during the initial inspection.

4.0 DATA DELIVERABLES

4.1 The Contractor shall provide data IAW CDRL A001; and

4.2 The Contractor shall provide data IAW CDRL A002.

5.0 GOVERNMENT-FURNISHED PROPERTY/MATERIAL/INFORMATION (GFP/GFM/GFI)

The Government does not anticipate any government furnished property/materials/information for this effort.

6.0 SECURITY

All work under this SOW is Unclassified.

7.0 TRAVEL

The Contractor will be required to travel in performance of this contract. The number of trips and types of personnel traveling shall be limited to the minimum required to accomplish work requirements and shall be coordinated with the technical point of contact. All travel shall be conducted in accordance with the Federal Travel Regulations (FTR).

8.0 POINT OF CONTACT (POC)

[redacted]

[redacted]

[redacted]

[redacted]